

Co-28Cr-6Mo Alloy Annealed-1st DP
Co-28Cr-6Mo Alloy Hot Worked-1st DP
Co-28Cr-6Mo Alloy Warm Worked-1st DP

Table 1.1.1 Typical mechanical properties of implant metals. [1]

Material	ASTM designation	Condition	Young's modulus (GPa)	Yield strength (MPa)	Tensile strength (MPa)	Fatigue endurance limit (at 10^7 cycles, $R = -1^c$) (MPa)
Stainless steel	F745	Annealed	190	221	483	221–280
	F55, F56, F138, F139	Annealed	190	331	586	241–276
		30% Cold worked	190	792	930	310–448
		Cold forged	190	1213	1351	820
Co–Cr alloys	F75	As-cast/annealed	210	448–517	655–889	207–310
		P/M HIP ^b	253	841	1277	725–950
	F799	Hot forged	210	896–1200	1399–1586	600–896
	F90	Annealed	210	448–648	951–1220	Not available
		44% Cold worked	210	1606	1896	586
		Hot forged	232	965–1000	1206	500
	F562	Cold worked, aged	232	1500	1795	689–793 (axial tension $R = 0.05$, 30 Hz)
Ti alloys	F67	30% Cold-worked Grade 4	110	485	760	300
	F136	Forged annealed	116	896	965	620
		Forged, heat treated	116	1034	1103	620–689

^bP/M HIP; Powder metallurgy product, hot-isostatically pressed.

^c R is defined as $\sigma_{\min}/\sigma_{\max}$.